UNNAMED 2026 GUIDE
P510/1 (56).

Marking codes
EX
AN
AP

ITEM – 1 **MODULE 10**

(Particles).

$At = A_0e^{-\lambda t}$, $\lambda = \frac{In^2}{5730}$ s^{-1}

$500 = 4000e^{\frac{In^2}{5730}t}$ **t = 17190 years**

Scores

Since **17190 years > 16900 years**, then the site should be preserved as part of the region's cultural heritage.

Wavelength of x rays

ITEM 1:
AN – 10 scores
EX – 12 scores
AP – 04 scores
26 scores

from eV $= \frac{hc}{Amin}$ $\lambda min, = \frac{6.6 \times 10^{-34} \times 3 \times 10^{8})}{1.6 \times 10^{-19} \times 80000}$

$\lambda_{min} = 1.55 \times 10^{-10}$ m

05

Since 1.55×10^{-11} m $< 1.6 \times 10^{-11}$ m, the scanner provides accurate information.

Production of X-Rays

- Diagram
- When low voltage heats the filament cathode, electrons are emitted thermionically.
- Emitted electrons are accelerated by high voltage across the tube towards the metal target.
- On hitting the metal target, 99% of the kinetic energy of electrons is converted into heat and 1% of its kinetic energy is converted into x-rays.
- For continuous production of x-rays, excess heat generated at the metal target is conducted by the anode to the cooling fins where it is extracted.

05

b) A line spectrum is used to identify a metal because each metal emits / absorbs light at its own set of wavelengths.

- The produced x rays are directed to the metal penetrate deep and dislodges the electrons in the inner ~~most~~ shells.
- ~~The excited~~ electrons jump to higher energy levels and ~~on their way to~~ when electrons transition to lower energy levels they emit wavelengths.
- This emitted light is passed through a prism / diffraction grating in a spectroscope producing a line spectrum.
- The positions of the spectral lines are compared with the known spectra of different metals.
- If the lines match those of a known metal, the metal is identified.

07

The dislodged

b)

1. Detecting broken bones
2. Testing welds, detecting cracks in metal
3. Determining time of death
4. To track how drugs move in the body

Explain any 2

- The photographic film is placed behind the suspected part of the body.
- X-rays are then directed to the suspected part where x-rays can pass through the broken part but are absorbed by unbroken bones.

Or

Key words in the scenario requiring the learners response
- Surprised.
- worried.
- wondered.
- surprised.
- Excited.
- Doubts.
- Needed to know.
- Didn't know.
etc.
- For devices only purpose or information

Setting
- Application
- Devices
- Explanations

observation
measurements.

ITEM 1

Code		weight 4	weight 3	weight 2	weight 1	
PC	BY	(9–12)	(7–8)	(4–6)	(0–3)	maximum scores 12
	AM	////	(7–10)	(4–6)	(0–3)	maximum scores 10
PA	AP	////	//	(3–4)	(0–2)	Maximum score 4

use: $\frac{3}{4}, \frac{2}{4}, \frac{1}{4}$ (the $\frac{1}{4}$ rule)

use $\frac{2}{3}, \frac{1}{3}$ (the $\frac{1}{3}$ rule)

$\frac{1}{2}$ (the $\frac{1}{2}$ rule)

NB: But we chose to follow scores (the usual ordinary marking) than the weighting scoring.

THE ELEMENT OF CONSTRUCT CALLED PARTICLES (P)

For Item 1 (compulsory); has 3 topics-

1.— Atomic particles ← cathode rays, Cathode Ray Tube (CRT), the CRO, Bain bridge mass spectrometer, J.J thomson's expt, mass defect, B.E, R-E/N, unified atomic mass

2 - Quantum theory ← ~~Bohr~~ Photo electric effect, Bohr, X-rays - Bragg's law & Bragg's law.

3 - Nuclear process ← Fusion, Fission / nuclear power plants.
- Radioactivity
- Devices

- The solid unbroken part will appear light but the broken parts will appear dark.

4)

- The drug is injected into the body, and inside the body it decays by emitting γ-rays (gamma-rays).
- Detectors around the body catch both gamma-rays at the same time, and the computer maps where they came from. Bright spots indicate where the drug has accumulated.

02

Item - 2 (Forces & motion).

a) Being Pushed Sideways

- As the car moved round the bend along a circular path, it continuously changes direction which means it undergoes centripetal acceleration directed towards the center. Aisha and peter`s bodies possess inertia and naturally tend to maintain their original straight-line motion.
- And inside a turning car the passengers experience an outward ~~friction~~ force known as the centrifugal.

04

Why was the car not sliding?

- The static friction between the car tyre`s and road surface is strong enough to provide the required inward centripetal force.
- The passengers slide sideways relative to the car until the lateral support of the seats or doors applies an adequate inward normal force to pull them along the curve.

03

$$F_c \le F_f,\ F_c = \frac{m \cdot \mu^2}{r},\ F_f = \mu \cdot m \cdot g$$

Did Jose negotiate the corner at the right speed?

Given constants:
r = 80 m, μ = 0.28, g = 9.8 ms⁻²

$$(m \cdot v_{max}^2) / r \le \mu mg \Rightarrow v_{max} = \sqrt{(\mu r g)}$$

Jose negotiating the corner at right speed?

r=80m, M = 0.28, g=9.81ms^{-1}

$\frac{MV^2_{max}}{r} \le \mu mg, V_{max} = \sqrt{\mu r g}$

$V_{max} = \sqrt{0.28 \times 9.81 \times 80} = 14.82 ms^{-1}$

06

Engineers recommend only 90% of this speed. V = 0.9x14.82 = 13.34ms^{-1} since Jose was travelling at 12ms^{-1} thus is below 13.34ms^{-1} he was negotiating the corner safely.

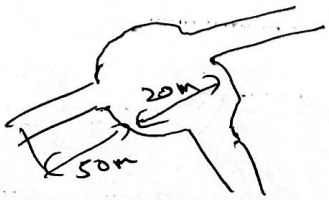
~~Page 4~~

Jose's Options at the Junction

Distance to clear junction: s = 50 + 20 = 70 m (he requires 70 m to be out of the junction, the other for safe).
Time before red: t = 4 s

Option 1: Accelerate at 1.4 ms⁻²

$s = u \cdot t + \frac{1}{2} \cdot a \cdot t^2$



NB:

Topics in Item 2

1 — Dimensions
2 — statics
3 — Linear motion
4 — motion under gravity
5 — solid friction.
6 — mechanical properties of materials
7 — Fluid mechanics. — Bernoulli's
8 — work, power & Energy.
9 — circular motion
- centripetal force
- centrifugal force
- conical pendulum
- vehicle — on flat road
- on banked road.

10 — Gravitation & satellites.

Setting $\left(\frac{2}{3} \times 10\right)$ in one scenario

i.e 7 topics.

4 — 7 topics.

$s = 12(4) + \frac{1}{2}(1.4)(4)^2 = 59.2$ m

He requires 70 m. So 59.2 m < 70 m, he does not clear the junction, he will be stranded right in the intersection when traffic light turns red.

Option 2: Decelerate at 0.4 ms^{-2}

$s = 12(4) - \frac{1}{2}(0.4)(4)^2 = 44.8$ m

44.8 m < 50 m, the car will remain entirely behind the stop line when light turns red. Hence, he will safely avoid entering the intersection during red light.

Final Conclusion: option 2 is significantly safer option, and option 1 guarantees a dangerous traffic violation and potential collision inside the junction.

b) Safety Challenges on Bends & Junctions

- The wet road hazard; during rainy periods, water acts as a lubricant between the tyre`s and the road surface drastically lowering coefficient of friction which reduces the maximum speed safe speed, causing normal drivers moving at standard speeds to lose traction and skid off road.
- Unbanked design vulnerability; rely entirely on tyre`s road friction for centripetal force is fundamentally flawed because friction is highly unpredictably and weather dependent.

Technical Advice to the Authorities

To eliminate the black sport and stabilize the flow of traffic, the following engineering modifications are advised.

- Redesign the bend with mechanical banking, the authorities should rebuild the sharp bend by tilting the road surface inwards at a strategic angle. By banking the curve, the normal reaction force splits into horizontal component of normal reaction that points inwards the center of the turn which minimizes skidding accidents during wet periods.
- Apply high-friction surface treatment; Resurface the bend using highly durable coarse aggregates embedded in a polymer. This artificial texturing maintains a high coefficient of friction even when wet, preventing vehicles from sliding out. (Tarmac the road)

widen the junction to increase radius of curvature

Item - 3

a) Variation in Sound Intensity and Clarity of a Public Address System

- Variation in loudness is due to the interference of sound waves coming from the two loudspeakers.
- At some positions, sound waves meet in phase, overlap, and undergo constructive interference, producing a loud sound.
- At other positions, they arrive out of phase, cancel each other out, and undergo destructive interference. Producing soft / low sound.

Calculation Context: Isaac counted 9 loud sound positions over a specific distance.

Distance walked: $d = v \times t = 3 \times 8 = 24$ m

Spacing between adjacent loud positions:

$Dx = \frac{24}{8} = 3.0$ m

~~$Dx = \frac{v}{n}\lambda = 3 \times 2 = 6$m~~

$\frac{1}{2}\lambda = \overline{AA} = 3$m

$\lambda = 3 \times 2$

$\lambda = 6$ m

a) Variation in sound intensity:

This is due to formation of stationary waves, the two loud speakers produce two progressive waves of the same amplitude, wavelength/speed/frequency, moving in opposite directions which meet & superpose forming stationary waves.

P.T.O

NB:

The Element of content of Energy (Items 3 & 4)-

1. Reflection of light
2. Refraction of light
3. optical ~~instruction~~ instruments.
4. Thermometry
   - Thermometers
     - digital infared thermometer
     - digital clinical thermometer.
   - Pyrometer.
5. Quantities of Heat.
6. Heat transfer.
7. Behaviour of gases.
8. Thermodynamics
   - Refrigerator | air conditioner
   - Carnot engine.
   - First law & second law of Thermodynamics.
9. SHM + Oscillations
10. Progressive waves:
    - wave equation
    - Interference
    - Diffraction
    - Polarisation.
11. Stationary waves + Sound waves (Pipes + strings)
12. Beats, Doppler effect ~~Determine~~ velocity of blood in the body (Resonance).

- At one point they meet in phase resulting into reinforcement forming Antinodes (A) hence loud sound
- Then out of phase cancellation forming Nodes (N), and low sound results

NB: Here, 5-7 topics are set, two items are set using a parallel way (i.e the two items have different topics)

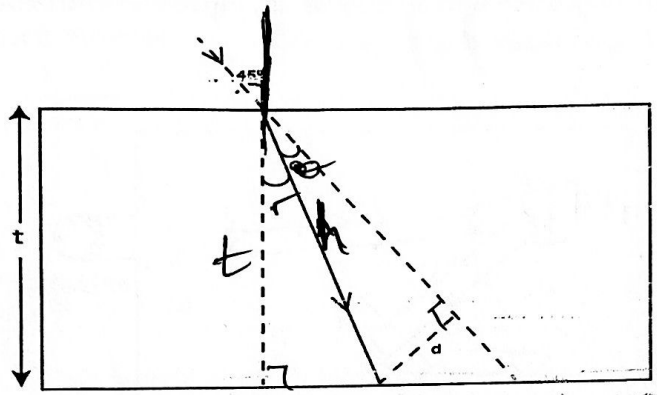
continuation of item 3.

$f = \frac{v}{\lambda} = \frac{340}{6} = 56.7 Hz$

Conclusion: Since 56.7 Hz > 55 Hz,

The public address system would be clear because the amplifier frequency is above the required minimum threshold of 55 Hz.

b) Lateral Displacement Through Glass



Using Snell's Law nsini = constant:

$1.0 \times \sin(45°) = 1.50 \times \sin(r)$

$r = \sin^{-1} \frac{\sin 45°}{1.5} = 28.1°$

$\cos(r) = \frac{t}{h}$ $h = \frac{t}{\cos r}$ --- (1)

$\sin(\theta) = \frac{d}{h}$, $h = \frac{d}{\sin\theta}$ --- (2)

$\frac{t}{\cos 28} = \frac{d}{\sin 16.9^0}$ (16.9°)

$\frac{1.2}{\cos 28.1} = \frac{d}{\sin 16.9}$

$d = 0.39545$ cm

$d = 0.4$ cm

Since 0.4 cm < 0.5 cm, the lateral displacement is strictly below the allowable limit. Therefore, the close-up photographs will maintain proper alignment.

Heating of a Pump Barrel and Bursting of a Volleyball

Why the pump barrel became hot:

- This is due to adiabatic compression and the thermodynamic work done on the system.
- When air is rapidly pumped into the ball, mechanical work is done directly on the air molecules, which significantly increases their internal energy and raises the temperature. The metal barrel conducts this thermal energy, making it hot to the touch.

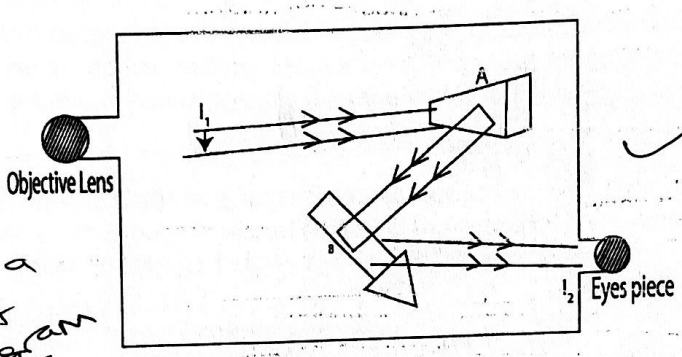
03

Why the volleyball burst in hot sunshine:

- In hot sunshine, the temperature of the air trapped inside the ball increases.
- This causes an increase in the average kinetic energy and rate of collisions of the air molecules, leading to a sharp rise in internal pressure.
- Simultaneously, the structural rubber material becomes structurally weaker and more pliable when exposed to intense heat. When the rising internal pressure exceeds the structural strength limit of the rubber, the ball bursts.

b) Optical Instrument Recommendation

Recommendation: Use Binoculars or a Refraction Telescope to effectively magnify distant athletes. (Name of device)



Redraw a correct diagram

Principles of Operation:

- The objective lens refracts light rays coming from a distant object to form a real, inverted image (I_1) at its principal focus in front of Prism A.
- Rays from I_1 incident on Prism A are totally internally reflected and turned around horizontally by the prism structure.
- On reaching Prism B, the rays undergo total internal reflection again but are flipped vertically, forming a final erect image (I_2) right-side up relative to the distant object, viewed through the eyepiece at infinity.

Item 4

- A concave mirror reflects parallel incoming solar rays precisely onto its focal point.
- The concave mirror tracking alignment is altered to face directly towards the sun.
- A blackened cooking pot is moved ~~incrementally~~ along the principal axis of the mirror layout.
- The positions correspond to an optimal focal setup adjusted until the reflected light forms the smallest and brightest concentrated spot directly on the container.
- The blackened cooking pot absorbs the radiation reflected from the mirror determining If the system is efficient for use.

solar power incident on the mirror: $A = \pi r^2$; $r = 1.25\,m$

$$P = IA\ ,\ A = \pi r_m^2\ ;\ r_m = 1.25\ m$$

$$P = 227 \times \pi \times (1.25)^2 = 1359.721\ W$$

40% reaches the blackened cooking pot:

Properties of thermal radiation
- travels at a speed of light of 3.0×10^8 m/s in a vacuum
- travel in straight lines
- reflected, refracted,

$$P_0 = 0.4 \times 1359.721 = 543.89\ W$$

Heat needed to raise the temperature for water to be used in biology practical:

$$543.89 \times 27 \times 60 = 3.5 \times 4200 \times (\theta - 20)$$

$$\theta = 79.74°C\ (= 80°C)$$

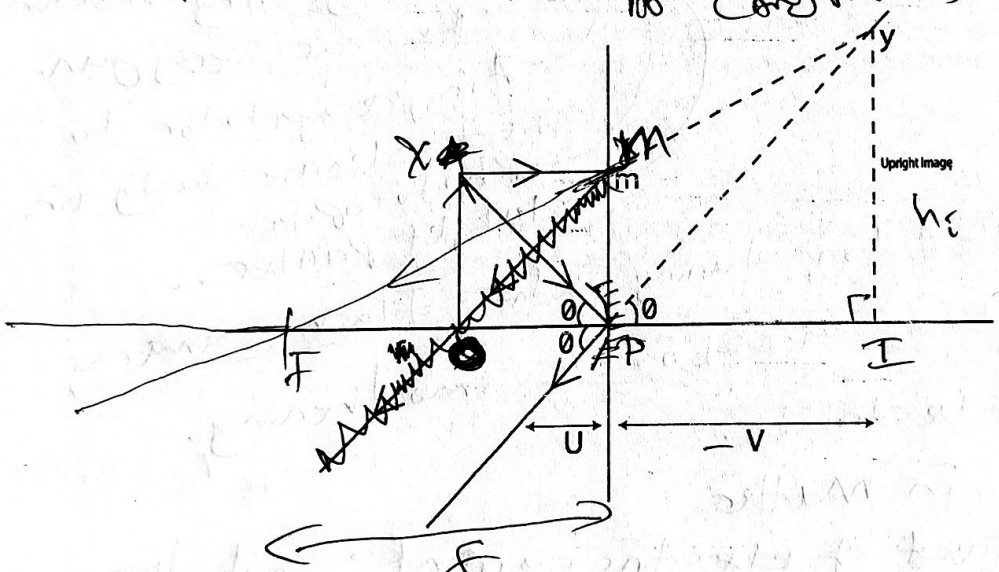
This matches the precise required baseline temperature needed for the biology practice.

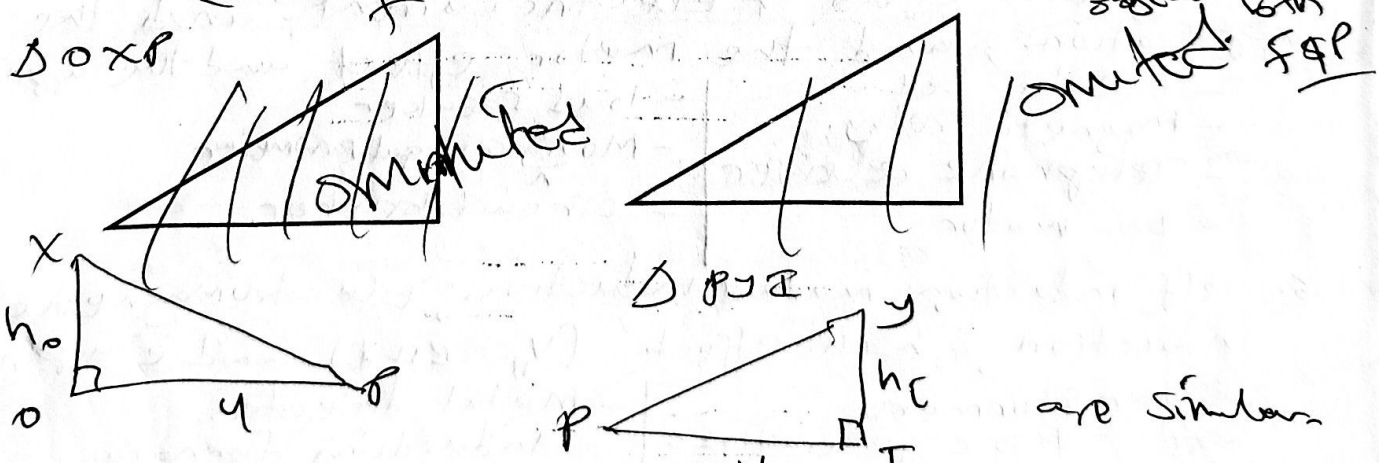
Technical Enhancements to Achieve Boiling Point (100°C)

- Increase effective area captures more solar power.
- Use a transparent cover creates greenhouse effect, reduces convection and radiation loses
- Add reflective side walls: Increases the effective collection aperture area.
- Use an active tracking mechanism: Keeps the reflector pointed at the sun for longer periods, maximizing daily average power output.

b)

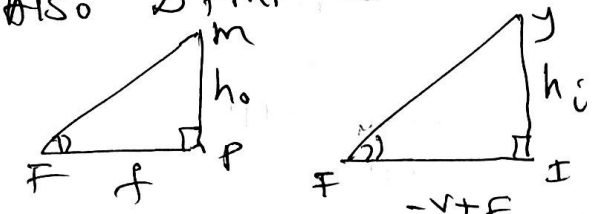
- A dentist requires a larger, clearer view of a small, hard-to-access tooth surface.
- When the tooth is placed between the focal point (F) and the pole (P) of a concave mirror, the image becomes enlarged, upright and virtual





$$\frac{y_I}{OX} = \frac{-V}{u} = \frac{h_i}{-h_o} \quad (1)$$

Also ΔFMP and ΔFYI are similar.



$$\frac{YI}{PM} = \frac{h_i}{h_o} = \frac{FI}{FP} = \frac{-v+f}{f}$$ ✓ AP

$$\frac{-v}{u} = \frac{-v+f}{f}$$

$-uf = -uv + uf$

dividing through by uvf

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ ✓ AP

NB: You can also select other 2 pairs of similar triangles

The 4th Element of construct (charges + fields) (Items 5 & 6):

1- Electrostatics and its applications:
- The Van de graaf generator
- The lightning conductor
- The car spray gun.
- Air precipitator to clearse dusty air.
- Photo copier.
- Inkjet printer.

2. Capacitors:- Action; tuning radios, Flash cameras, a.c to d.c smoothening.

3. Current Electricity.

4- Magnetism in Matter.

5, Magnetic effect of electric current; and the applications; and the motor effect and the applications

- Electric bell	- loud speaker
- Magnetic relay.	- Moving coil galvanometer
- Telephone receiver	- circuit breaker.
- D.C motor	

6. Self induction, mutual induction, Electromagnetic induction, hall effect ($V_H = BIL/d$) and its applns.

- Transformers.	- Metal detectors
~~- A.c & D.c generators.~~	- Induction furnace
- Ignition coil	- A.c & d.c generators.

7- A.c circuits & its application.

8- Digital electronics, semiconductors, logic gates, and truth tables and applications.

$\frac{yl}{mp} = \frac{x+v}{f}$ since $ox = mp$

$\frac{yl}{OX} = \frac{u-v}{f}$

$\Rightarrow \frac{-v}{u} = \frac{-v+u}{fv}$, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

~~Item 5~~ ITEM 5:

a) why electricity transmitted at 135KV but supplied to homes at about 240V

- Electrical power is transmitted at very high V=135KV, so that the current is small
- since P=VI for the same power, increasing voltage reduces current and power lost in transmission lines is Ploss = I^2R, the smaller current greatly reduces heat and energy losses in transmission wires

Why are bluish glows and faint sparks

$P=VI$, $I = \frac{5X10^6}{135000} = 37.04A$

Since 37.04A < 40A bluish faint light is caused by a large current flowing in wires

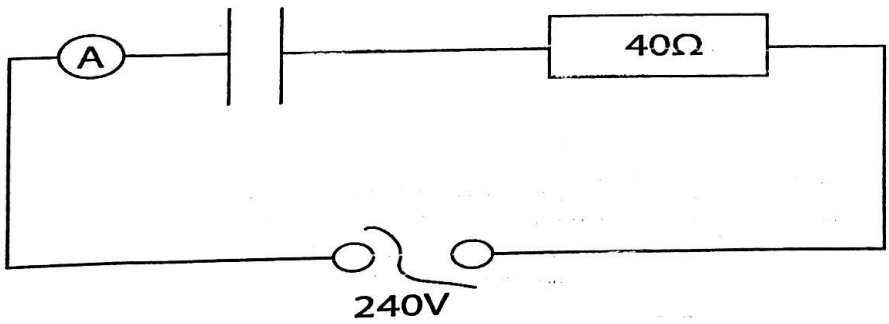
the glow is due to Corona discharge the high voltage produces a very high electric field intensity around the conductor

Air molecules near the wire become ionized, the ionized air emits a faint bluish light and sometimes produces a hissing sound not caused by a large current flowing in wires since transmission current is limited < 40A, power loss is kept very low.

Concern of health center technician

$\frac{V_s}{V_P} = \frac{N_s}{N_P}$, $V_s = \frac{240 \times 25}{500} = 12V$

Transformer appropriate because it steps down the main supply from 240V to 12V medical monitoring circuit is to be connected to a fuse device, 40A or 6.0A



$V = ZI$

$Z = \sqrt{X_C^2 + R^2}$, $X_C = \frac{1}{2\pi fc}$

$X_C = \frac{1}{2\pi \times 106 \times 10^{-6} \times 50} = 30.029\Omega$

$Z = \sqrt{(30.029)^2 + 40^2} = 50.018\Omega$

Action of a step down transformer (OA scores)

- Step down transformer

connected to A-C voltage in the primary coil. Producing alternating current

This causes a changing magnetic flux in the soft iron core which links with the secondary coil of less turns by mutual induction.

Hence increasing emf of 240V.

$240 = 50.018\ I,\ I = 4.80A$

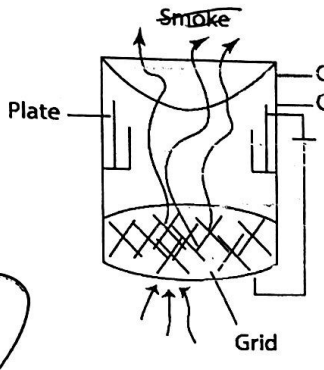
4.0A ~~40A~~ fuse; Too small it will blow immediately because 4.80A > 4.0A

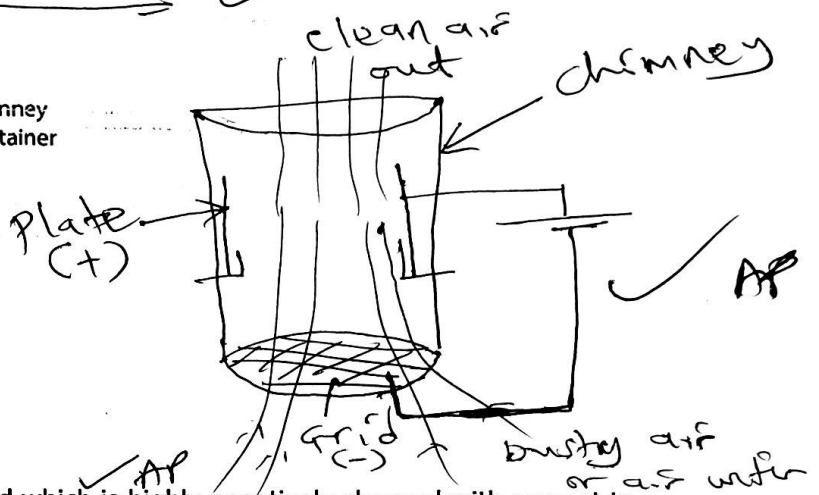
Page 12 (First Sheet)

6.0A fuse; can carry 4.80A safely and will not blow if there is a fault / overheating
∴ The 6.0A fuse is most appropriate

b) Recommend an electrostatic

- ~~based~~ solution; electrostatic air ~~precipitation.~~





- In a chimney there is a grid which is highly negatively charged with respect to the plates
- As smoke passes through the grid its particles are negatively charged
- when the ~~smoke~~ reaches the regions of plates, the particles are attracted to the plates
- Hence ~~smoke~~ gets out of the chimney free of particles
- The particles accumulating on the plates falls off in the container when heavy enough.

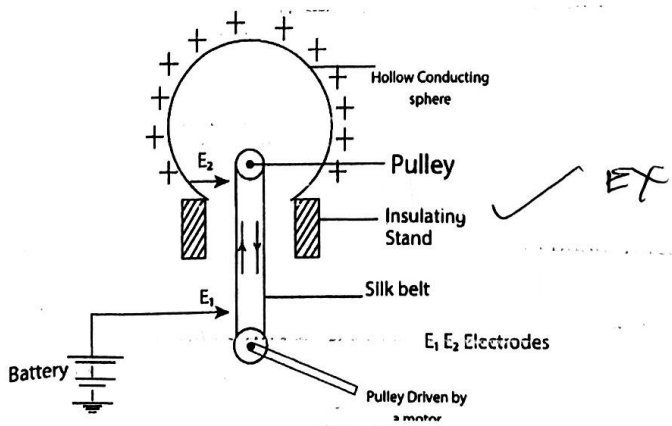
Advantages

- Reduces respiratory health risks to patients and staff
- Improves air quality in the intensive care unit.
- Has high efficiency
- Removes very fine dust particles

omitted

item 6

Work of a van de Graaff generator



- Electrode E_1 is made positive relative to the Earth, at E_1 there is high charge density and high electric field intensity. Air around E_1 gets ionized
- Positive ions are repelled to the belt which drives them up to the sphere
- At the sphere negative charge is induced at E_2 and positive charge induced at the surface of the sphere
- Point action takes place at E_2, negative ions are repelled to the belt and discharge it before it moves over the pulley.
- The process continues until the charge on the sphere can ionize the air around it. Potential of the sphere is about 1×10^6 V relative to Earth.

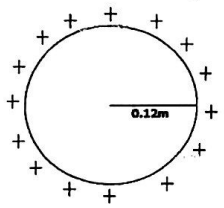
Decision to accept the generator or not,

Energy supplied by the motor, $E = V I t$

Page 14

$E = 12 \times 0.5 \times 60 = 360J$ (in put) OR

Energy stored by the sphere.



$Q = 2.4 \times 10^{-6}\ C$

$C = 4\pi \varepsilon_0 r$

$C = 4\pi (8.85 \times 10^{-12}) \times 0.12$

$C = 1.33 \times 10^{-11}\ F$

Electrostatic energy (output) OR $V = \frac{Q}{4\pi\varepsilon_0 r}$ Then $E = \frac{1}{2}CV^2$ OR $\frac{1}{2}QV = E$

$E = \frac{Q^2}{2C}$

$E = \frac{(2.4 \times 10^{-6})^2}{2 \times 1.33 \times 10^{-11}} = 0.216J$

Efficiency of conversion of energy

$\eta = \frac{0.216}{360} \times 100\% = 0.06\%$

Required efficiency of energy conversion is 0.06% > 0.05%, so the generator should be accepted.

For the van de Graaff generator to be accepted, the sphere must be smooth, spherical and clean, thus charge must be uniformly distributed over the surface. To check for this

- A proof plane is placed on the surface of the dome that is charged, and then transferred into a metal can placed on the cap of a neutral gold leaf electroscope.
- Divergence on the gold leaf electroscope is noted.

- The experiment is repeated with the proof plane placed at different parts of the dome and corresponding divergencies noted
- If the divergencies are equal, the charge is then uniformly distributed, since efficiency of energy conversion is greater than 0.05%, and charge distributed uniformly over the dome, the generator can be accepted.

b) - As the van de Graaf is kept running charge on the dome kept on increasing until a value when the insulation of air breaks down.

since $Q = E_0 \times 4\pi \varepsilon_0 r^2$

$Q = 3.0 \times 10^6 \times 4\pi (8.85 \times 10^{-12}) (0.12)^2$

$Q = 4.8 \times 10^{-6}\ C$

$CV = Q$

$V = \frac{Q}{4\pi\varepsilon_0 r}$

Intensity $= \frac{kQ}{r^2}$

$E = \frac{Q}{4\pi\varepsilon_0 r^2}$

$E \times 4\pi\varepsilon_0 r^2 = Q$

When the charge increases to 4.8×10^{-6} the electric field intensity is high enough to break down insulation of the air around the sphere thus causing sparks around the dome.

END

OR $E = \frac{V}{r} \Rightarrow V_{max} = E_{max} r$

$= 3.6 \times 10^6 \times 0.12$

$= 4.32 \times 10^5\ V$

$Q_{max} = C V_{max}$

$= 1.33 \times 10^{-11} \times 4.32 \times 10^5$

$= 5.748 \times 10^{-6}\ C$ Appd.

P510/1

PHYSICS

Paper 1

(Theory)

AUGUST 2026

3 HOURS

UNNASE MOCK EXAMINATIONS

Uganda Advanced Certificate of Education

PHYSICS

Paper 1

3 hours

INSTRUCTIONS TO CANDIDATES

*This examination paper consists of two sections; **A** and **B**. It has **six** items.*

*Section **A** has **two compulsory** items.*

*Section **B** has two parts; **I** and **II**. Respond to only **one** item from **each** part.*

*Respond to **four** items in all.*

*Any additional item(s) responded to, will **not** be scored.*

*Responses to **each** item should start on a **fresh page**.*

Silent non-programmable scientific calculators may be used.

SECTION A.

Respond to **all** *the items in this section.*

ITEM 1

During the construction of a lakeside fish landing site near Lake Kyoga, workers uncovered an ancient canoe buried in wet soil and recovered a piece of wood and a metal artifact from the site. An Archaeologist who was working at the site predicted that the same place could have been used as a landing site before but could not know the period when this could have been. Laboratory analysis showed that the recovered piece of wood had a carbon-14 activity of 500 counts per minute, while a recent sample of the same species of the same mass have an activity of 4000 counts per minute. The half-life of carbon-14 is 5730 years. Regulations are that the site would be preserved as part of the region's cultural heritage if it had been a landing site over 16900 years ago.
To investigate the metal artifact without damaging it, scientists used a handheld X-ray fluorescence scanner whose X-ray tube operated at 80KV. For the scanner to provide accurate information, the minimum wave length of the x-rays used should not exceed $1.6 \times 10^{-11}m$. The scanner identified the artifact as copper after detecting characteristic X-ray spectral lines emitted from the metal but the workers were not convinced whether the scanner gave accurate results.
The workers at the site wondered how X-rays are produced, and how characteristic spectral lines detected by the scanner help in identifying nature of the Metal.
John one of the workers who had some knowledge in physics became curious on how the either of the physics concepts used in the investigation at the site could be put to other use in real life.

Support material:

- Plancks Constant h = 6.6×10^{-34}Js
- Speed of light = 3×10^{8} ms^{-1}

Task:

As a physics student,

(a) Analyze the scenario to establish whether the site was preserved, and address all the concerns of the site workers.

(b) Examine a detailed response to John.

ITEM 2

Frequent accidents at sharp corners and road junctions in a certain city have become a major road safety concern. Authorities have installed traffic lights and enforced speed limit regulations on roads identified as accident hot spots. Jose was driving a car with two passengers, Aisha and Peter along a flat circular road of radius 80m before joining a straight road leading to a junction. As the car negotiated the circular road, Aisha and Peter felt as if they were being pushed and sliding sideways. Peter and Aisha wondered why the car was not sliding like them and whether Joel was negotiating the corner at the right speed.
Traffic engineers recommended that vehicles should not exceed 90%of

the maximum speed that can be negotiated without skidding around the corner. Many drivers have complained that vehicles often fail to negotiate the sharp corner when driving at high speed, especially during wet conditions. The government is considering improving the corner to allow vehicles to negotiate it safely at high speed. The authorities hope that this would help to improve safety on the roads and would need technical advice.

After leaving the circular road, Jose was 50m from the near edge of a junction that was 20m wide and was driving at a speed of $12ms^{-1}$ when the traffic light changed from green to orange. The orange light would remain on for 4.0 seconds before turning red. Jose considered either accelerating uniformly at $1.4\ ms^{-2}$ or decelerating at $0.4ms^{-2}$. He wanted to be completely out of the junction before the light turned Red.

Hint: total mass of the car and its occupants was 1,200kg

Coefficient of friction between the road and the tyers was 0.28.

Support material:

- Acceleration due to gravity g = 9.81 ms^{-2}

TASKS:

(a) Analyze the information to address Peter and Aisha's worries, and evaluate Jose's options at the junction.

(b) Assess the safety challenges and give technical advice to the Authorities.

SECTION B

PART 1.

Respond to **one** *item from this part*

ITEM 3

A secondary school organized its annual sports day and used several technologies to improve communication, photography, and sports equipment management. Two loudspeakers connected to the same amplifier were placed 100 m apart on opposite sides of the playground. As Isaac, the school entertainment minister, stood at the midpoint between the speakers, he heard a loud sound. He then walked steadily along the line joining the speakers at a speed of 3.0 m s^{-1} and, during the next 8.0 s, counted 9 loud sound positions, including the one at the midpoint. Isaac did not understand why there was variation in sound intensity at different positions in the sports field. The public address system would be clear if the frequency of the amplifier is at least 55Hz.

At the prize-giving ceremony, the championship trophy was displayed inside a glass case to protect it from damage. A photographer viewed the engraved school name through a flat glass wall of thickness 1.2 cm and refractive index 1.50. A ray from the engraving struck the glass at an angle of incidence of 45°.

The photographer was informed that if the lateral displacement of the ray exceeded 0.50 cm, the engraved name could appear noticeably shifted in

close-up photographs and this would distort the photograph alignment. Isaac would like to know if the close-up photographs of the trophy would have proper alignment.
Meanwhile, before the volleyball finals, the games teacher used a hand pump to inflate the match balls. After repeated pumping of air into the balls, he noticed that the metal barrel of the pump became hot. By the afternoon, several balls had become unusually hard and one volleyball burst while players were warming up in the hot sunshine. Isaac did not understand what caused the heating of the Burrel of the pump or what caused busting of the ball.
During the athletics event, the guest of honour was seated at the far end of the playground a considerable distance from the starting point of the races. He informed the headteacher that he was finding if difficult to see athletes at the start of each race. The school has a variety of optical instruments, the headteacher needed recommendation on which optical instrument can help the guest of honour and explanation on how it works. Hint: speed of sound in air is $340ms^{-1}$

Task:

(a) Analyze the information to provide response to Isaac, and determine whether the public address system would be clear
(b) Provide suitable recommendation to the head teacher.

Item 4

In one of the rural schools in Uganda, their learners were tasked to develop a solar water heating system to provide warm water for laboratory use, especially for Biology practical's.
They are using a large concave mirror of surface diameter 2.5 m to concentrate sunlight onto a blackened cooking pot of efficiency 40%, used to boil 3.5 kg of water from 20°C using solar radiation of intensity of 277 W m^{-2} incident onto a mirror.
During the first trial, the water became only slightly warm, the learners suspected that the pot is not placed at the correct position where sunlight is most concentrated and had no idea on how to overcome this also became suspicious of the properties of radiation incident onto the mirror. After positioning the pot, it took students 27 minutes to obtain a reasonable temperature raise of about 80°C for effective use in the laboratory for biology practical. And when learners continued with the heating system water never reached 100°C and this left them confused on what improvements necessary to achieve water boiling at 100°C.
Asiimwe one of the students thereafter inquired from their teacher if there are other real-life applications of concave mirrors. The teacher informed him that the same mirrors are used by dentists for teeth examination. The teacher challenged him to research on how this application can be illustrated using a ray diagram and how it can be used to establish the relationship between object, image distances from the mirror and its focal length.

Support material:

- Specific heat capacity of water = 4200 Jkg^{-1} K^{-1}

Task:

a) Help students with guidelines to properly position the container for optional light intensity reception and analyze if the system would be efficient for continuous use and respond to students' worries.

b) Examine Asiimwe`s concern.

SECTION B.

PART II

Respond to **one** *item from this part*

ITEM 5

A rural health center and a nearby trading center were recently connected to the national electricity grid. During the official launch of the project, the project engineer reported that the electric power is transmitted from the generating station at 135KV and that the transmission line delivers a power of $5.0 \times 10^6 W$.The member of parliament wondered why electricity is transmitted at high voltage yet only a bout 240V of electric power is needed for home consumption. He could not understand how this high voltage would be reduced and needed explanation about the device that would be used.

During the trial period before launch, residents reported seeing faint sparks and bluish glow around transmission line fittings on humid evenings. This worried the residents who started speculating that the cause of this was very high current flowing through the transmission wires. The engineer assured the residents that the current flowing through the wires is safe and he expects it to cause very minimal energy loss and for minimal power loss, the current through transmission wires should not exceed 40A. He rather pointed to ionization as the cause of what residents were observing. Residents could not understand this and needed more explanation.

The health center uses an x-ray machine supplied through a transformer whose primary coil has 500 turns and the secondary coil has 25 turns. The transformer is connected to a 240V ac mains supply, the ratting of the x-ray tube indicates 12V, but the hospital technician is not sure if the provided transformer is appropriate for use.

A medical monitoring device in the health center operates from the same 240V, 50Hz AC supply and contains a 40Ω connected in series with a capacitor of $106\mu F$. The medical monitoring circuit is to be connected to a fuse devise, a 4.0A fuse and a 6.0A fuse are available but the technician needs to pick the most appropriate one for use.

At the health center, the intensive care department often becomes dusty during dry season because of heavy traffic on the nearby road. The health workers reported that dust accumulation was affecting air quality

in the intensive care unit. The health center management need a suitable method for removing fine dust particles from the air before it enters the building.

Task: As a physics student,

(a) Analyze the information and address the worries of the member of parliament, the residents and the health center technician.

(b) Assess the situation and recommend an electrostatics-based solution for improving air quality at the intensive care department.

ITEM 6

A certain school has been experiencing a challenge of being supplied fake school laboratory equipment. The school procurement committee has come up with a regulation of testing any equipment supplied to the school before it is accepted. A small ~~Van de~~ Graaff generator was supplied to the school; the generator uses a motor driven insulating belt to transfer charge to a spherical metal dome of radius 0.12m. The school laboratory technician did not know the system works and even what to check for. According to the operating manual, charge of $2.4 \times 10^{-6}C$ can be obtained on the dome if the motor driving the generator belt operates at 12V and draws a current of 0.5A for one minute, and at least 0.05% of the electrical energy should be stored as electrostatic energy. The spherical done acts as an isolated spherical capacitor.
Donald a senior six physics student was requested to work with the laboratory t echnician. Donald suggested that they can test for charge distribution and how much of the motor electrical energy is converted to stored electrostatic energy. During the test, the generator is kept running for more than one minute. At some time, sparks were observed around the dome. This surprised both Donald and the technician. Before the two could complete the test, Donald was called for his next lesson.

Support material:

Permittivity of free space $\varepsilon_0 = 8.85 \times 10^{-12} Fm^{-1}$

Dielectric strength of air $= 3.0 \times 10^{6} Vm^{-1}$; or NC^{-1} (Intensity)

TASK:

(a) As a physics student, analyses the information, and help the laboratory technician to understand the working of the system and make a decision on whether to accept the generator or not.

(b) Explain what could have caused the sparks around the dome, with clear details to convince Donald.

END